

GRIDER FIELD, AR, USA

WMO: 723417

Lat: 34.180N

Lon: 91.934W

Elev: 63

StdP: 100.57

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93988

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.8	(c) -3.8	(d) -13.0	(e) 1.2	(f) -3.1	(g) -10.4	(h) 1.6	(i) -1.4	(j) 10.2	(k) 12.6	(l) 9.2	(m) 10.1	(n) 3.6	(o) 10	(p) 0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	36.2	25.3	34.9	25.2	33.7	24.9	26.9	33.1	26.4	32.5	25.8	31.7	3.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	29.7	24.8	19.9	29.3	24.1	19.1	28.7	84.7	33.1	82.2	32.5	80.0	32.0	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	8.0	7.2	-8.8	37.8	2.5	1.9	-10.7	39.1	-12.2	40.2	-13.6	41.2	-15.5	42.6
			-10.0	28.2	2.4	1.0	-11.7	28.9	-13.1	29.5	-14.5	30.0	-16.2	30.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	6.2	8.3	12.6	17.6	22.3	26.4	27.8	27.4	24.0	18.0	11.8	7.5
	DBStd	8.86	5.94	5.75	5.33	4.24	3.52	2.35	2.24	2.41	3.29	4.46	4.88	5.29
	HDD10.0	429	147	92	34	3	0	0	0	0	0	2	36	115
	HDD18.3	1526	377	283	187	62	10	0	0	0	3	63	203	338
	CDD10.0	3182	31	45	116	232	381	491	552	541	421	249	89	36
	CDD18.3	1237	2	2	11	41	132	241	293	282	173	52	6	1
	CDH23.3	12458	5	11	74	327	1117	2496	3262	3017	1623	487	36	3
	CDH26.7	5549	0	1	8	60	376	1138	1597	1511	707	149	3	0
Wind	WSAvg	3.0	3.6	3.7	3.8	3.8	3.2	2.6	2.2	2.0	2.3	2.7	3.0	3.3
Precipitation	PrecAvg	1292	103	108	132	123	128	90	97	84	88	106	111	132
	PrecMax	1842	211	268	349	415	330	222	328	203	261	357	344	382
	PrecMin	834	13	19	25	25	20	2	5	0	3	0	18	18
	PrecStd	259	54	57	64	86	76	56	58	52	53	77	69	80
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	25.0	27.8	30.2	33.0	36.2	37.6	38.6	36.2	32.0	26.8	23.4
		MCWB	17.8	18.2	18.8	20.9	23.2	24.6	26.1	25.5	22.8	21.4	19.3	18.7
	2%	DB	20.9	22.4	25.6	28.4	31.6	34.5	36.0	36.4	34.1	30.2	24.0	21.0
		MCWB	16.9	16.9	18.1	19.9	22.5	24.7	25.8	25.6	23.6	21.1	17.8	17.3
	5%	DB	18.3	20.1	23.3	27.0	30.3	33.2	34.7	34.9	32.6	28.2	22.2	18.5
		MCWB	15.3	16.0	16.9	19.3	22.2	24.4	25.7	25.3	23.3	20.1	17.6	15.7
	10%	DB	16.0	17.8	21.3	25.2	28.9	32.3	33.5	33.4	31.0	26.3	20.0	16.3
		MCWB	12.6	14.1	15.9	18.5	21.8	24.0	25.5	25.0	22.7	19.2	16.2	13.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	19.7	20.6	23.0	25.2	27.0	27.9	27.7	26.1	24.2	21.3	19.9
		MCDB	22.1	22.9	25.4	27.7	30.4	32.7	33.8	33.9	32.2	28.9	24.2	22.5
	2%	WB	17.8	18.4	19.4	21.7	24.1	26.0	27.1	26.8	25.2	23.1	19.8	18.2
		MCDB	20.0	21.2	23.8	26.3	29.0	31.7	33.2	33.3	31.0	27.5	22.6	20.1
	5%	WB	16.0	16.9	18.2	20.8	23.5	25.5	26.5	26.2	24.5	21.9	18.4	16.4
		MCDB	18.1	19.5	22.0	25.0	28.3	31.1	32.6	32.7	29.7	25.7	21.3	18.4
	10%	WB	13.1	14.7	16.8	19.8	22.7	24.9	26.0	25.7	23.8	20.7	16.8	14.0
		MCDB	15.2	17.3	20.3	23.7	27.4	30.4	31.8	31.8	28.6	24.3	19.4	15.9
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.0	10.7	11.3	10.5	10.3	10.2	10.9	11.8	12.3	10.9	9.4
		MCDBR	12.6	13.6	13.0	12.7	11.7	11.4	11.6	12.7	13.5	14.6	13.2	12.0
		MCWBR	9.2	9.1	7.3	6.1	4.7	4.0	4.1	4.2	4.7	6.6	8.3	9.0
	5% WB	MCDBR	10.9	11.8	11.3	10.4	10.1	10.2	10.5	11.4	11.0	11.0	11.1	10.7
		MCWBR	9.1	9.0	7.4	6.1	4.7	4.2	4.2	4.4	4.4	6.2	8.3	8.9
Clear-Sky Solar Irradiance	taub	0.322	0.332	0.361	0.402	0.434	0.463	0.492	0.486	0.419	0.361	0.338	0.318	
	taud	2.507	2.470	2.413	2.296	2.260	2.201	2.155	2.158	2.334	2.471	2.492	2.527	
	Ebn at Noon	889	917	913	886	856	828	801	799	843	873	862	872	
	Edh at Noon	84	97	111	131	137	145	151	148	119	96	85	78	
All-Sky Solar Radiation	RadAvg	2.53	2.99	4.20	5.27	5.81	6.49	6.28	5.78	4.96	3.90	2.80	2.23	
	RadStd	0.23	0.46	0.38	0.36	0.42	0.47	0.35	0.45	0.39	0.52	0.31	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page